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The directed genus of the de Bruijn graph

Alfred W. Hales[a,*], Nora Hartsfield[b]

[a] *Center for Communications Research, 4320 Westerra Court, San Diego, CA, 92121, USA*
[b] *Department of Mathematics, Western Washington University, Bellingham, WA 98225, USA*

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Abstract

Define the *directed genus*, $\Gamma(G)$, of an Eulerian digraph G to be the minimum value of p for which G has a 2-cell embedding in the orientable surface of genus p so that every face of the embedding is bounded by a directed circuit in G. The directed genus of the de Bruijn graph D_n is shown to be

$$\Gamma(D_n) = 2^{n-1} + 1 - \frac{1}{2(n+2)} \sum_{d|n+2} \phi(d) 2^{\frac{n+2}{d}}.$$

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1. Introduction

The *(binary) de Bruijn graph* D_n is the directed graph whose vertices are the binary n-tuples $\mathbf{v} = (v_0, v_1, \ldots, v_{n-1})$ and with edges from each $\mathbf{v}$ to $(v_1, v_2, \ldots, v_{n-1}, 0)$ and $(v_1, v_2, \ldots, v_{n-1}, 1)$. Hence D_n has 2^n vertices and 2^{n+1} edges, and edges correspond to $n+1$-tuples $(v_0, \ldots, v_{n-1}, v_n)$. Furthermore, D_n is Eulerian, with two edges entering and two edges leaving each vertex. An Eulerian circuit in D_{n-1} yields a Hamilton cycle in D_n, and is called a *de Bruijn sequence of span n*. For example, D_2 is shown in Fig. 1 and D_3 in Fig. 2.

We shall be considering surfaces S that are compact orientable 2-manifolds without boundary. Such a surface of genus p can be thought of as a sphere with p handles. A *directed embedding* of an Eulerian directed graph G into such a surface S is a 2-cell embedding (*i.e.* the interior of each face is homeomorphic to a disc) of G into S such that every face is bounded by a directed circuit in G. The *directed genus of* G, $\Gamma(G)$ is the minimum value of p for which G has a directed embedding into a surface of genus p (see [1]). For example, the embedding of D_3 in Fig. 2 is not a directed embedding in the plane; see Fig. 4.

The main result of this paper is the following theorem.

* Corresponding author.
E-mail addresses: hales@ccrwest.org (A.W. Hales), frog@cc.wwu.edu (N. Hartsfield).

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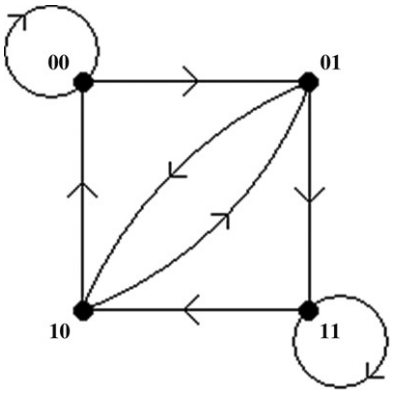


Fig. 1. Planar embedding of D_2.

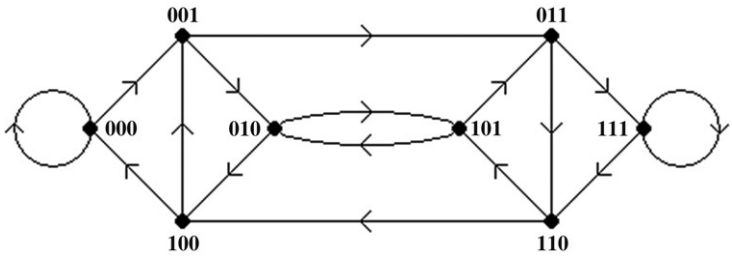


Fig. 2. Planar embedding of D_3.

Theorem 1. *The directed genus of the binary de Bruijn graph with 2^n vertices is*

$$\Gamma(D_n) = 2^{n-1} + 1 - \frac{1}{2(n+2)} \sum_{d|n+2} \phi(d) 2^{\frac{n+2}{d}}.$$

We introduce the notation

$$Z(n) = \frac{1}{n} \sum_{d|n} \phi(d) 2^{n/d}$$

where, by the so-called Burnside Lemma, $Z(n)$ counts the number of binary circular necklaces of length n up to rotation. Hence our theorem can be restated as

$$\Gamma(D_n) = 2^{n-1} + 1 - \frac{1}{2} Z(n+2).$$

We shall also give some related open problems and some applications of our results to the classification of de Bruijn sequences.

2. Background

Suppose that we are given a directed embedding of D_n into an orientable surface S. Choose a specific orientation of S. Then the neighborhood of (the image of) any vertex $\mathbf{v} = (v_0, v_1, \ldots, v_{n-1})$ must look like one of the two diagrams in Fig. 3 where we have labeled with $n+1$-tuples the edges in and out of $\mathbf{v}$.

Conversely, suppose that we are not given the embedding, but just a choice of one of these two diagrams for each vertex $\mathbf{v}$ of D_n. Then we claim that a directed embedding into an appropriate surface S is determined by this information. There will be two types of faces, *counterclockwise* faces and *clockwise* faces. Consider first the counterclockwise faces. Any edge will lie in exactly one of these, and if, for instance, $0\mathbf{v}$ is such an edge and diagram

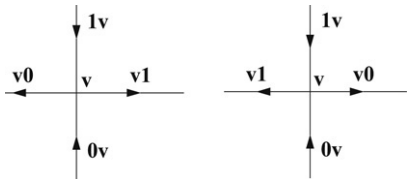


Fig. 3. Possible vertex neighborhoods.

(1) applies at **v**, then 0**v** must be followed by **v**0 along the face; similarly 1**v** must be followed by **v**1 along such a face. On the other hand, if diagram (2) applies, this is reversed; *i.e.* 0**v** is followed by **v**1 and 1**v** by **v**0. Similarly, each edge will lie in exactly one clockwise face and the diagram choices determine edge successions here as well. Hence all faces are determined, to be "glued together" along edges to give a (compact orientable) surface S, and we obtain a directed embedding of D_n into S. (Note that a face may intersect itself in one or more vertices, but not in an edge.)

These remarks enable us to work with appropriate diagram choices in analyzing directed embeddings of D_n.

3. Proof of the theorem

We first exhibit an embedding that achieves the genus given in the theorem. We choose the appropriate diagram for a vertex $\mathbf{v} = (v_0, v_1, \ldots, v_{n-1})$ to depend on the *parity* of the sum of the v_i; *i.e.* we choose diagram (1) if $\sum_{i=0}^{n-1} v_i$ is even and diagram (2) if $\sum_{i=0}^{n-1} v_i$ is odd. Consider now a counterclockwise face of the associated embedding. For **v** of even parity an edge $e\mathbf{v}$ will be followed by the edge $\mathbf{v}e$, and for **v** of odd parity $e\mathbf{v}$ will be followed by $\mathbf{v}\bar{e}$. Any pair of successive edges corresponds to an $n+2$-tuple and we see that this $n+2$-tuple is $e\mathbf{v}e$ for **v** of even parity and $e\mathbf{v}\bar{e}$ for **v** of odd parity. Hence the sum of the entries in any such $n+2$-tuple will be even. A similar argument shows that for a clockwise face the sum of the entries in any $n+2$-tuple given by successive edges will be odd.

Consider now three successive edges along any face. These determine two overlapping $n+2$-tuples, $(v_0, \ldots, v_{n+1})$ and $(v_1, \ldots, v_{n+2})$. The argument just given implies that v_0 must equal v_{n+2}. Hence the edges around any face are obtained by taking all cyclic rotations of an $n+2$-tuple. We can thus count faces by counting orbits of the set of $n+2$-tuples under cyclic rotation using the Burnside Lemma, obtaining that the number F of faces of the embedding is

$$Z(n+2) = \frac{1}{n+2} \sum_{d|n+2} \phi(d) 2^{\frac{n+2}{d}}.$$

There are $V = 2^n$ vertices and $E = 2^{n+1}$ edges in D_n, so applying the Euler–Poincare formula (see [1])

$$V - E + F = 2 - 2p$$

we solve for p to obtain

$$p = 2^{n-1} + 1 - \frac{1}{2} Z(n+2).$$

This establishes the upper bound for $\Gamma(D_n)$. Now for the converse, we show the lower bound. Suppose that we are given a 2-cell embedding of D_n into an orientable surface S. Considering as above pairs of successive edges along any face, we obtain from each face a cyclic sequence of (overlapping) $n+2$-tuples that give a decomposition of D_{n+2} into a collection of disjoint circuits. Golomb [3] conjectured in 1967 that no such decomposition could contain more than Z(n+2) circuits, the number obtained from cyclic rotations. This conjecture was proved by Mykkeltveit [4] in 1972, and he actually proved a somewhat stronger result (conjectured by Lempel in 1971), namely that the minimum number of vertices that can be removed from D_n, for any n, to leave a graph with no circuits is $Z(n)$. In any case, Mykkeltveit's result implies that for any 2-cell embedding of D_n into an orientable surface S we have that $F \leq Z(n+2)$. This implies, using Euler's formula as above, that the genus p of S satisfies

$$p \geq 2^{n-1} + 1 - \frac{1}{2} Z(n+2).$$

Hence we have completed the proof of Theorem 1, namely that the directed genus $\Gamma(D_n)$ is given by

$$\Gamma(D_n) = 2^{n-1} + 1 - \frac{1}{2}Z(n+2) = 2^{n-1} + 1 - \frac{1}{2(n+2)} \sum_{d|n+2} \phi(d) 2^{\frac{n+2}{d}}. \quad \square$$

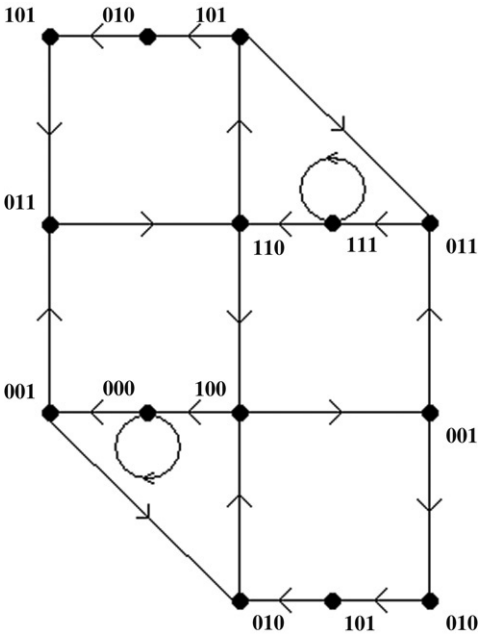


Fig. 4. Directed embedding of D_3 in a torus.

4. Open questions and applications

Suppose that we consider more general embeddings by dropping the "directed" restriction; *i.e.* we ask for the genus (in the usual sense) of D_n (see [5]). Already for $n = 3$ we see that this is a different question. Our formula gives that $\Gamma(D_3) = 1$, whereas Fig. 2 gives a (nondirected) embedding of D_3 in the plane. See Fig. 4 for a picture of D_3 directed embedded in the torus (the appropriate edge identifications should be made.)

This "is" the directed embedding given by our construction, although note that we have not shown that ours is the only directed embedding of D_n into a surface of genus $\Gamma(D_n)$ and in fact there may be several different embeddings that achieve the minimum.

We can also ask about the directed genus of the k-ary de Bruijn graph $D_{n,k}$. This is the Eulerian directed graph with vertices given by n-tuples from the set $\{0, \ldots, k-1\}$ of cardinality k, and with edges from $(v_0, v_1, \ldots, v_{n-1})$ to $(v_1, \ldots, v_{n-1}, v_n)$ as usual. Hence the number V of vertices is k^n and the number E of edges is k^{n+1}. Note that $D_{n,2}$ is just D_n and that $D_{1,k}$ is the complete directed graph on k vertices. The problem of determining the directed genus $\Gamma(D_{n,k})$ is open in general. We have determined this only along one "border" of the n, k square. Along the other "border", with $n = 1$, this turns out to be nothing but the ordinary genus of the complete graph on k vertices. But for arbitrary n, k the question appears to be wide open—and difficult. Although Mykkeltveit's theorem does generalize to the k-ary case, this is not what is needed for our question. All that one can get is a relatively weak upper bound on the number of faces F in a directed embedding of $D_{n,k}$.

For the directed genus of $D_{n,k}$ we have an upper bound of

$$\Gamma(D_{n,k}) \leq 1 + \frac{1}{2}k(k^n - k^{n-1}) - \frac{1}{2}\left(\frac{1}{n+2}\left(\frac{1}{k}\right) \sum_{d|n+2} \phi(d) k^{\frac{n+2}{d}} \max\{2, (d,k)\}\right).$$

This follows from a straightforward extension of our proof of Theorem 1, taking faces to correspond to $n+2$-tuples which sum to 0 or 1 mod k. It is not, unfortunately, a lower bound for all cases.

Our "canonical" embedding of D_n into a surface of genus $\Gamma(D_n)$ can be used in several ways to obtain invariants to assign to de Bruijn sequences. Recall that there are $2^{2^{n-1}-n}$ different de Bruijn sequences of span n and these are just Hamilton cycles in D_n or (equivalently) Eulerian circuits in D_{n-1}. An adequate classification of these sequences by invariants or otherwise has long been an important open problem (see for instance [2]).

Suppose that a de Bruijn sequence of span n is given. Thinking of this as a Hamilton cycle in D_n, we can associate to it a path on the surface S of genus $p = \Gamma(D_n)$ given by our embedding, and then take the *homology class* of this path. This gives an element of the abelian group Z^{2p} as an invariant for the de Bruijn sequence.

A second possibility, given a de Bruijn sequence or Hamilton cycle in D_n, is to *declare* this cycle to give a face for some directed embedding in some orientable surface S. Since the cycle visits each vertex, this will determine a choice of diagram (from Fig. 3) for each vertex, and hence will determine all faces for the embedding and indeed the surface S itself. We can then take as an invariant for the de Bruijn sequence the integer $g(S)$, the genus of the surface S.

The properties of these and other invariants remain to be investigated.

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